Technical record: boundaries, ground and limits

Version V2. Planar UTM19S figures are distinguished from geodesic machine metrics. Screening data is preliminary; image review is not engineering certification.

Test / capacity	Value / method
whole_site_km2	1657.8435417846936
master_km2	8.635
town_km2	5.76
dwellings	7872
residents	19680.0
jobs	2150
housing_blocks	99
school_places	3100
building_fixed_overlap_m2	0.0
building_channel_screen_overlap_m2	0.0
building_flood_screen_overlap_m2	0.0
base_gap_m2	0.0
base_overlap_m2	1.1920928955078125e-06
land_partition_gap_m2	0.0

Exclusive territorial account

Gross area and current district edge logic; fixed context is unchanged and is not counted again as new development.

District / primary use	Area km2	Edge / basis
D00-F / Fixed agrivoltaic pilot / fixed_context	13.892	Exact issued fixed boundary; unchanged
D00-L / Fixed living pilot / fixed_context	2.481	Exact issued fixed boundary; no internal redesign
D07 / Mountain stewardship / ecology	269.723	Source DEM steep/high terrain; preliminary conservation study, not statutory exclusion
D01 / Southwest settlement and farm services / mixed_use	113.246	SW access basin; northeastern edge is a declared urban service limit, not a built-up boundary
D02 / Western dry agriculture research / agriculture	424.315	Western fan and seasonal drainage system; testing sites only after water/soil verification
D03 / Northern energy research / solar_agriculture	213.049	North shoulder north of proposed research access; no installed full-district PV claim
D04 / Central production and transfer / industry	251.934	Transfer-corridor operating basin; no residential capacity credited
D05 / Eastern gateway and logistics / logistics	291.975	Eastern access/transfer study outside mountain reserve; external connection unsecured
D06 / Dry landscape and field-service links / reserve	77.228	Remaining seasonal-drainage links and unresolved terrain; explicit landscape/research role, not a lake

Complete construction capacity account

New / cumulative capacities, from current building inventory; household ratio 2.5 is assumed. Regional research districts receive no invented resident or job capacity.

Phase	Homes new / total	Residents new / total	Jobs new / total	Objective
1	1776 / 1776	4440 / 4440	905 / 905	Complete western neighbourhood, civic core, utilities and first productive halls
2	2112 / 3888	5280 / 9720	805 / 1710	Eastern living district and additional productive capacity
3	3984 / 7872	9960 / 19680	440 / 2150	Northern neighbourhoods and final staged halls

Per-zone detail is in phasing_programme.json and the phased geometry. P1 has one 750-place school for an assumed 666 pupils; P2 has 1,500 for 1,458; P3 has 3,100 for 2,952. This arithmetic does not establish service staffing, age mix, school funding, employment demand or a lawful settlement entitlement.

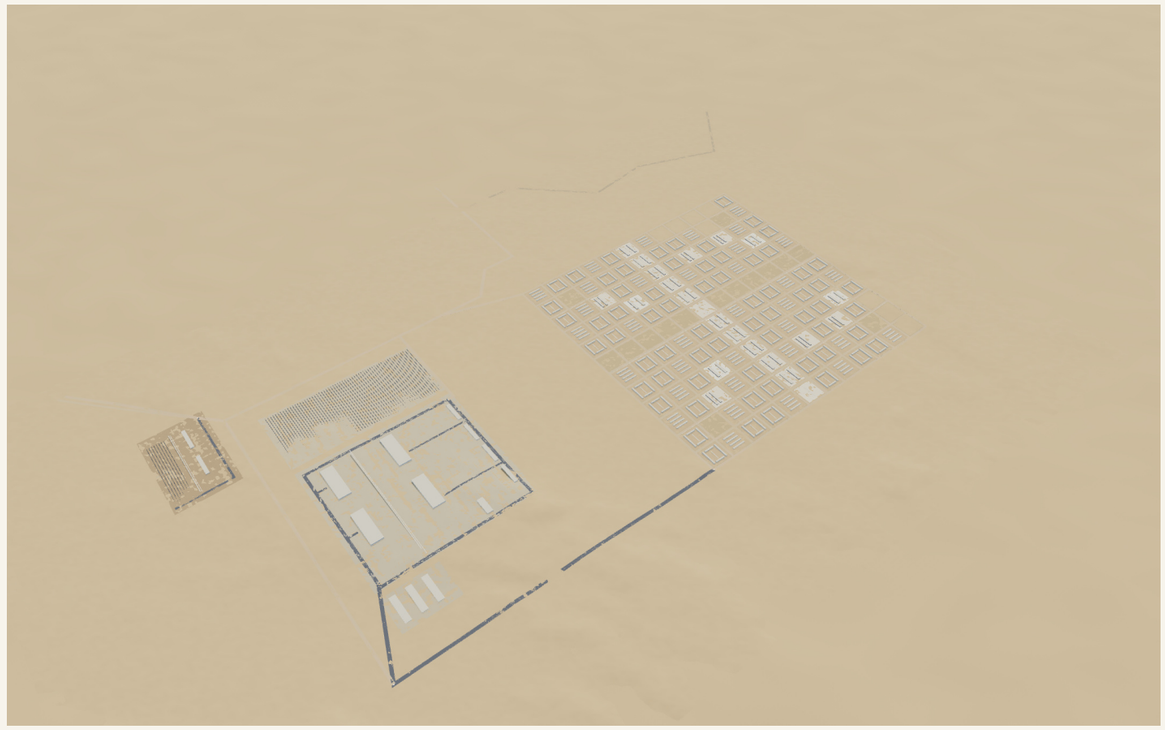
Resource scenarios: water and power

Recomputable arithmetic; fixed-pilot consumption and any future irrigation are additional and unknown.

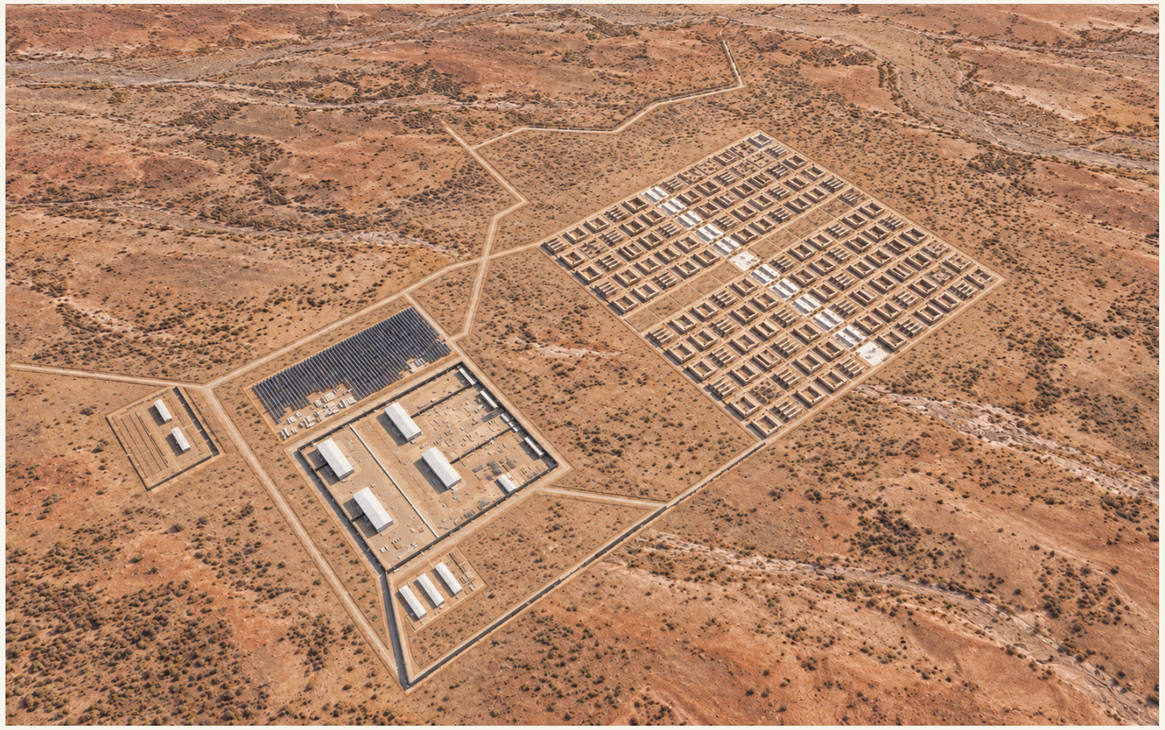
Scenario variable	Value
domestic_gross_m3y	610572.0
civic_gross_m3y	107748.0
work_gross_m3y	22575.0
compute_makeup_m3y	9110.4
gross_use_m3y	750005.4
reuse_m3y	100000
fresh_abstraction_m3y	706527.6086956522
distribution_loss_fraction	0.08
full_it_mw	8
pue_assumed	1.3
solar_mwp	26.784
solar_yield_kwh_kwp_y	1800
solar_mwh_y	48211.2
compute_mwh_y	91104.0
fixed_pilot_water	Additional and unknown; no supply entitlement confirmed
trial_irrigation	Not included; zero crop/irrigation capacity credited until lawful budget and soil trials

city-overall / control and finished image

Complete southwest settlement system / full build-out / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

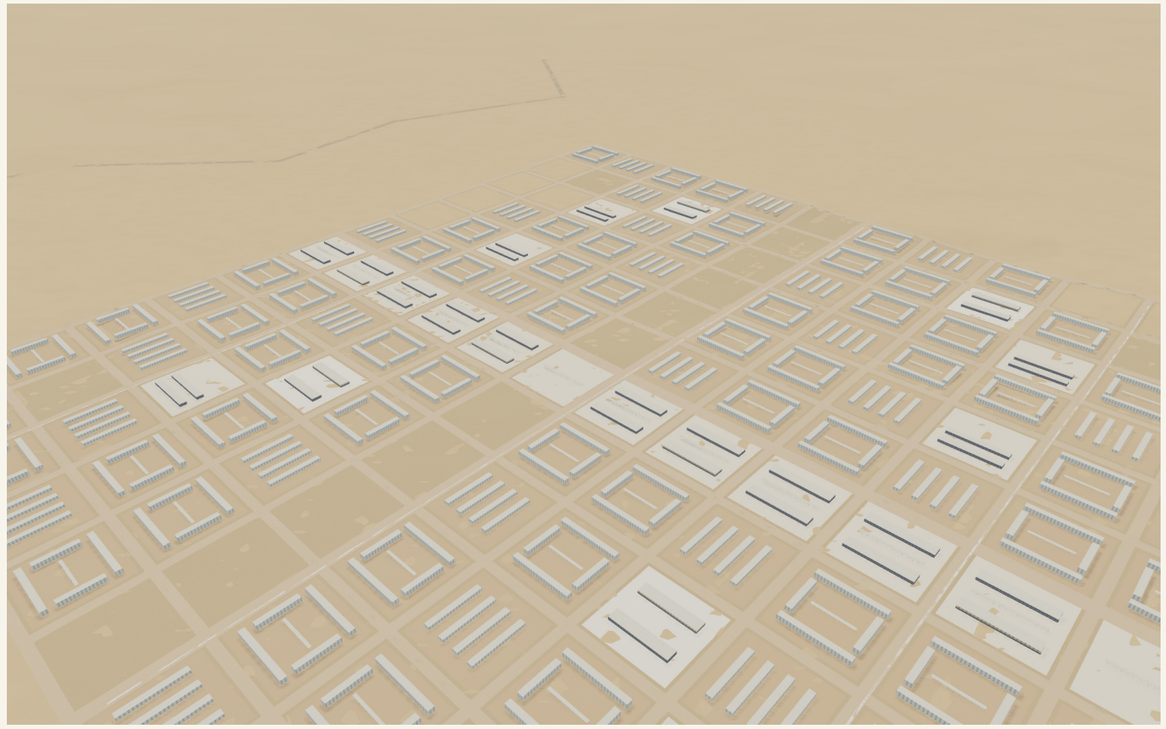


SELECTED FINISHED CONCEPT

Camera UTM19S: [578300, 6551200, 5100]; target [574400, 6554200, 0]; FOV 44 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

city-centre / control and finished image

Town civic band and living districts / local aerial / Full build-out scenario; node core enabled in Phase 1





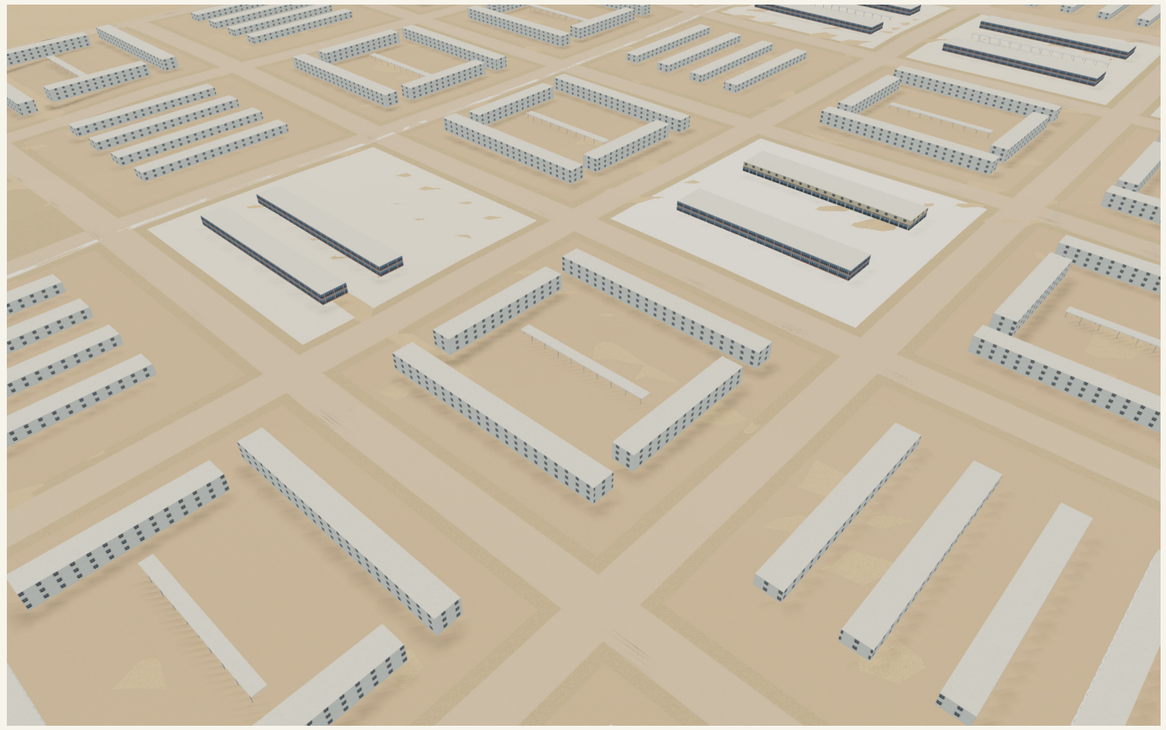
ORIGINAL GEOMETRY CONTROL

SELECTED FINISHED CONCEPT

Camera UTM19S: [575600, 6554300, 1050]; target [574500, 6555470, 0]; FOV 48 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

A01-aerial / control and finished image

A01 open-court housing / local aerial / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

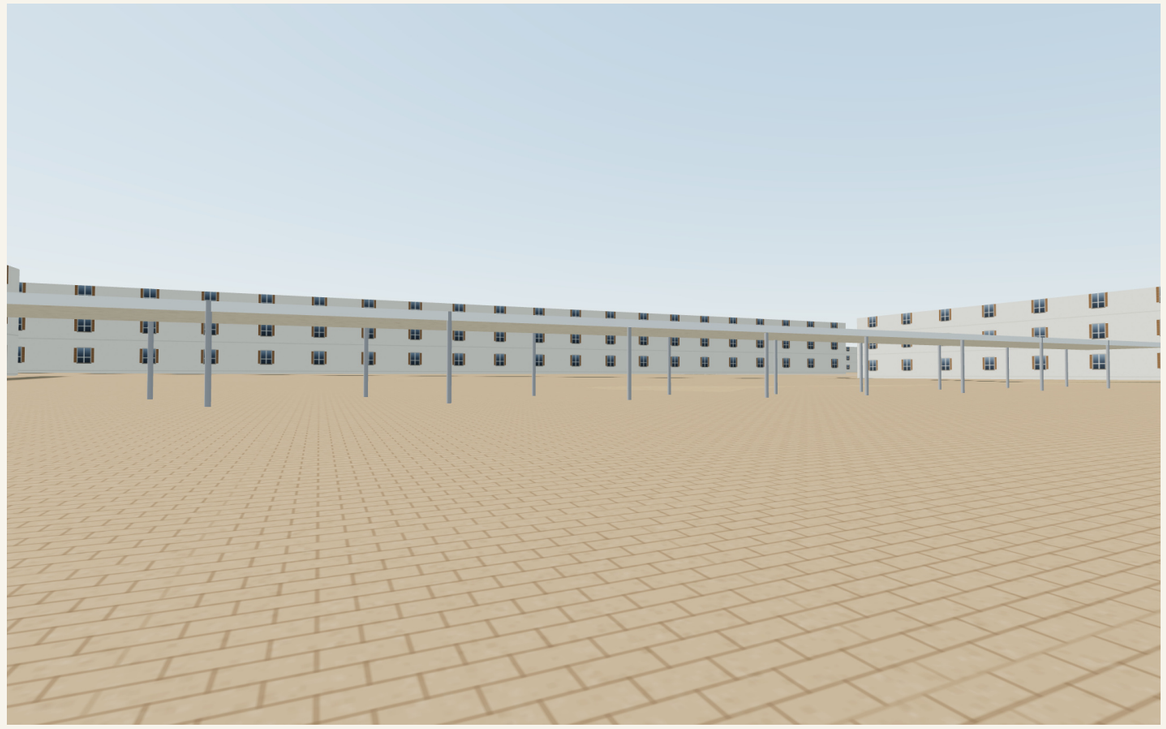


SELECTED FINISHED CONCEPT

Camera UTM19S: [574319.9999999997, 6554560.000000003, 260]; target [574099.9999999997, 6554800.000000003, 0]; FOV 46 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

A01-eye / control and finished image

A01 shared court / eye-level / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

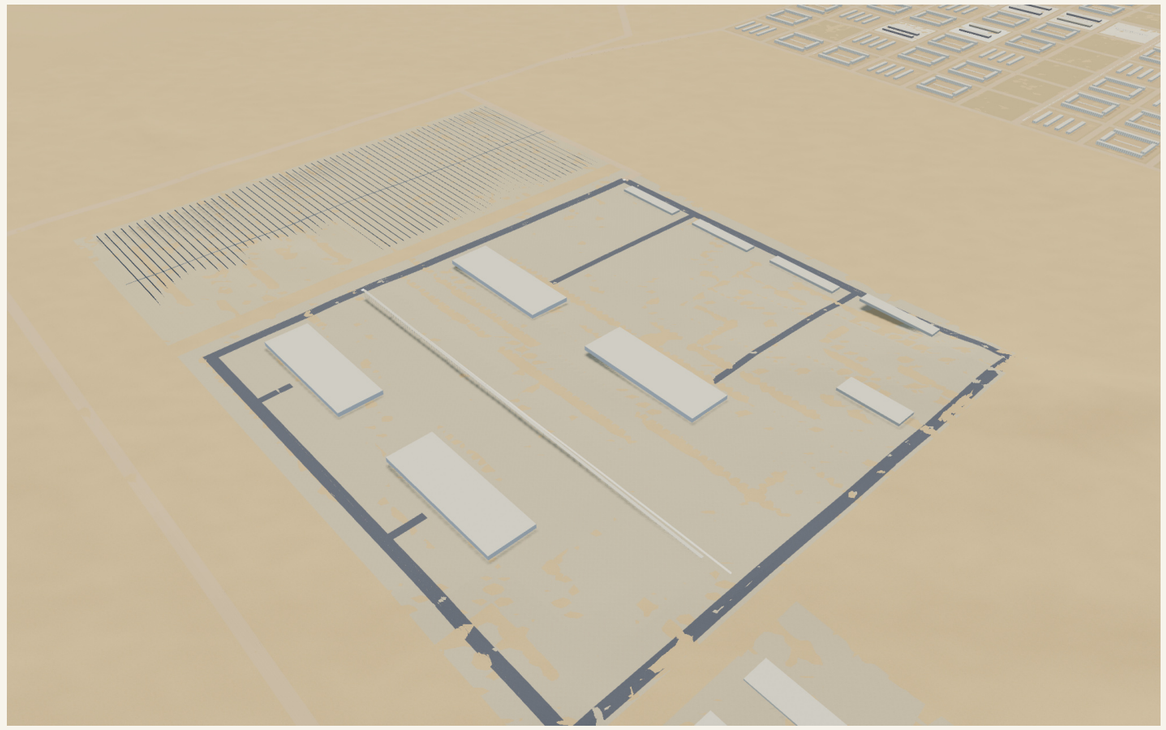


SELECTED FINISHED CONCEPT

Camera UTM19S: [574071.9999999997, 6554770.000000003, 1.7]; target [574107.9999999997, 6554850.000000003, 2]; FOV 63 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

B03-aerial / control and finished image

B03 four staged halls and repair / local aerial / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

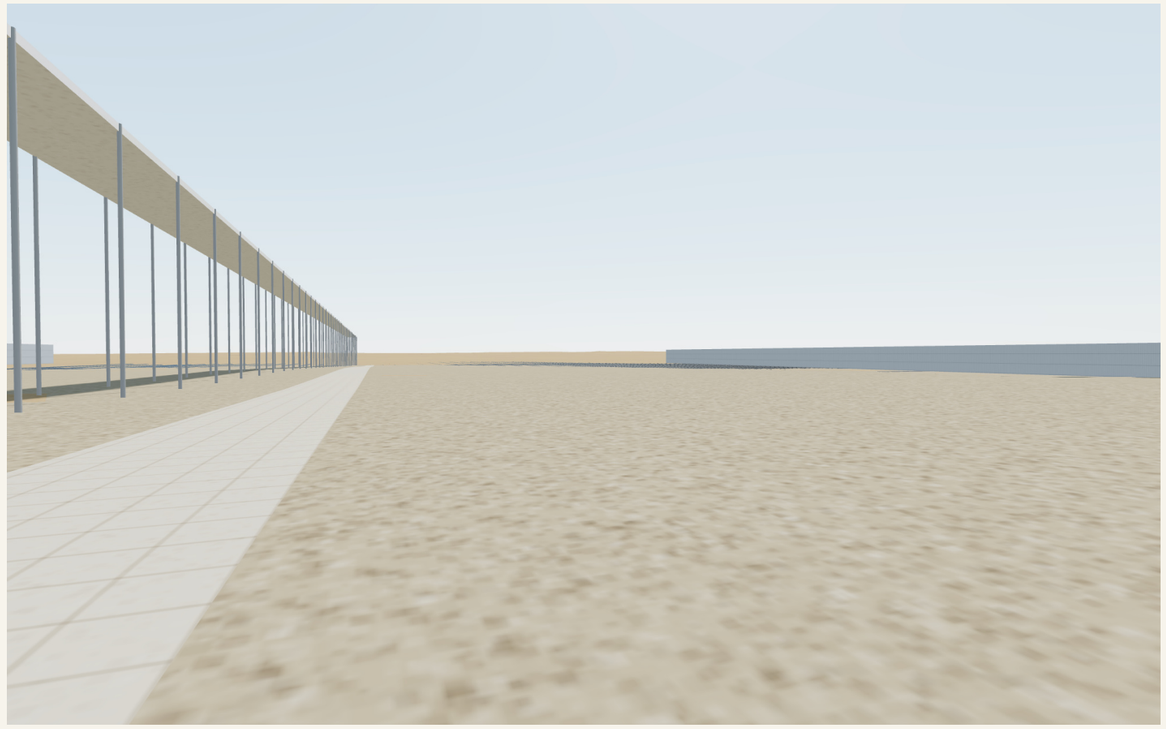


SELECTED FINISHED CONCEPT

Camera UTM19S: [575650, 6551550, 1150]; target [574600, 6552500, 0]; FOV 44 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

B03-eye / control and finished image

B03 protected worker spine / eye-level / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

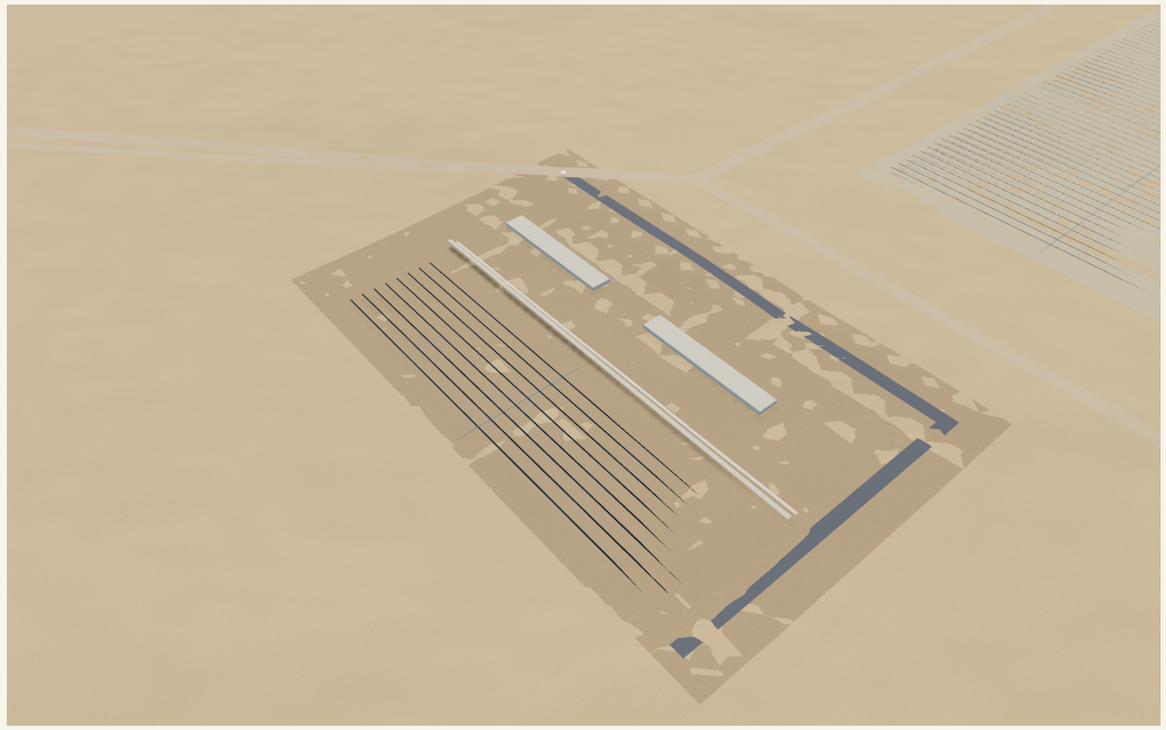


SELECTED FINISHED CONCEPT

Camera UTM19S: [574500, 6552318, 1.7]; target [574300, 6552390, 3]; FOV 64 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

C02-aerial / control and finished image

C02 new trial field and learning edge / local aerial / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

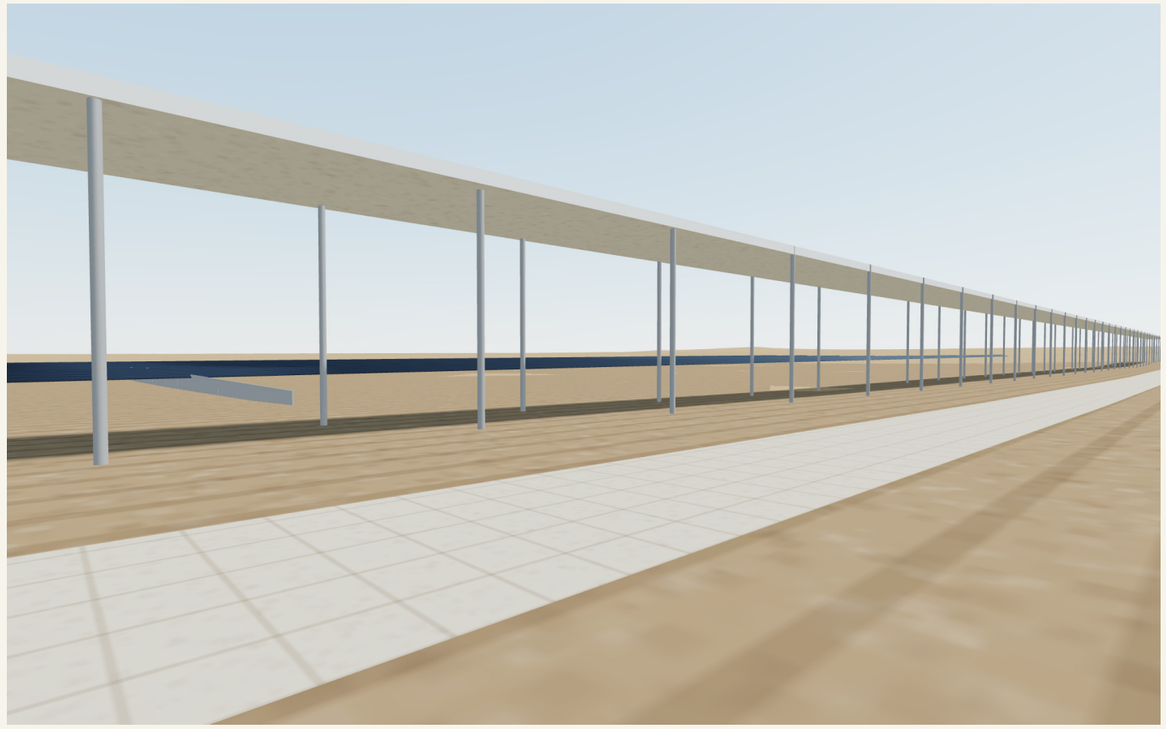


SELECTED FINISHED CONCEPT

Camera UTM19S: [574000, 6550800, 750]; target [573400, 6551300, 0]; FOV 45 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

C02-eye / control and finished image

C02 public shade edge facing new trial rows / eye-level / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL



SELECTED FINISHED CONCEPT

Camera UTM19S: [573410, 6551330, 1.7]; target [573290, 6551200, 2]; FOV 63 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

P1 operations and the 30 percent water stress test

Phase-one gross demand and the proposed public-service rules; all operating inputs remain declared scenarios.

P1 water use	m3/year / assumed basis
Domestic + civic	162,060 / 4,440 people x100L/day
Compute makeup	4,555.2 / 4MW IT, PUE1.3, WUE0.1L/kWh
Industry + panel cleaning	10,502.5 / worker use plus1,000 panel cleaning
Gross total	177,117.7 / fixed-pilot consumption additional unknown
30 percent shortfall	53,135.31 / remaining123,982.39

Water stress sequence. Halt new occupation. First stop nonessential panel cleaning(1,000), compute makeup(4,555.2) and nonessential work use(7,330.5). Halving civic use saves12,154.5; the remaining28,095.11 reduction would reduce the domestic allowance from85 to about67.7L/person/day. Essential-service allocation and recovery must be independently verified before restrictions; normal public life is not assumed to continue.

Firm obligations. The proposed public/cooperative utility owns essential interfaces; compute is a replaceable concession with a decommissioning bond. Publish daily aggregate meters, monthly accounts and a quarterly independent audit. Household traces stay private. P1 assumes an unsecured12MW grid connection;0hours storage and0% hourly clean matching are credited. Occupancy requires actual income, utility and service-funding evidence.

Source record and visual review

Public package: original model, cameras, controls, prompts, selected illustrations and dated review remain available separately.

Issued sources. Scope and fixed pilots: brief/site-package/boundaries and pilot-area. Terrain: Copernicus DEM GLO-30 source package, with its supplied licence and credits. Channels/flood bands: organiser DEM-derived studies, not surveyed limits. External roads: supplied OpenStreetMap extract; no on-site national/provincial road is asserted.

Retained original work. The earlier Everyday City terrain-cost regional corridor study and steep/high terrain study are retained with attribution to reviews/withdrawn-v1. No other entry's urban layout is copied. All town, housing, campus, farm-edge and phase geometry is new for V2.

Visual review. Eight distinct geometry cameras support two city views and aerial/eye-level pairs at A01, B03 and C02. The first overall finish invented planted trees and was rejected; the selected revision removes them. Controls never substitute for finished public views. Canopy members, access, building gaps, four compute halls and two farm sheds have been visually compared; this is not a pixel-level or construction certification.

Remaining verification. Survey, detailed grading, noise/dust and thermal impacts, accessible routes, junction/crossing design, canopy engineering, resource rights, hourly grid/storage modelling, school operation and demand/employment evidence. Status remains an authorised public preview; automatic checks and content review are reported separately.

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